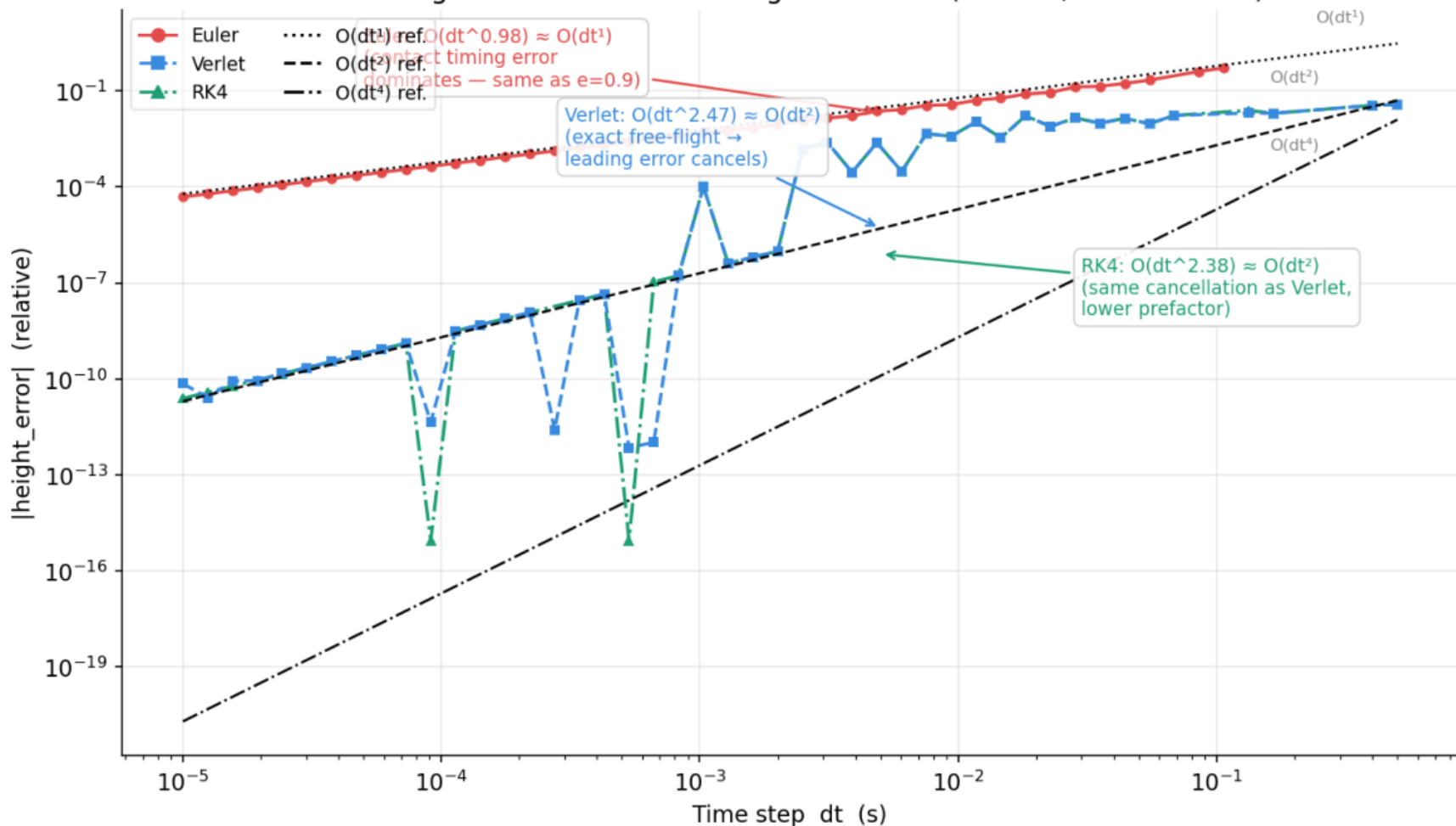
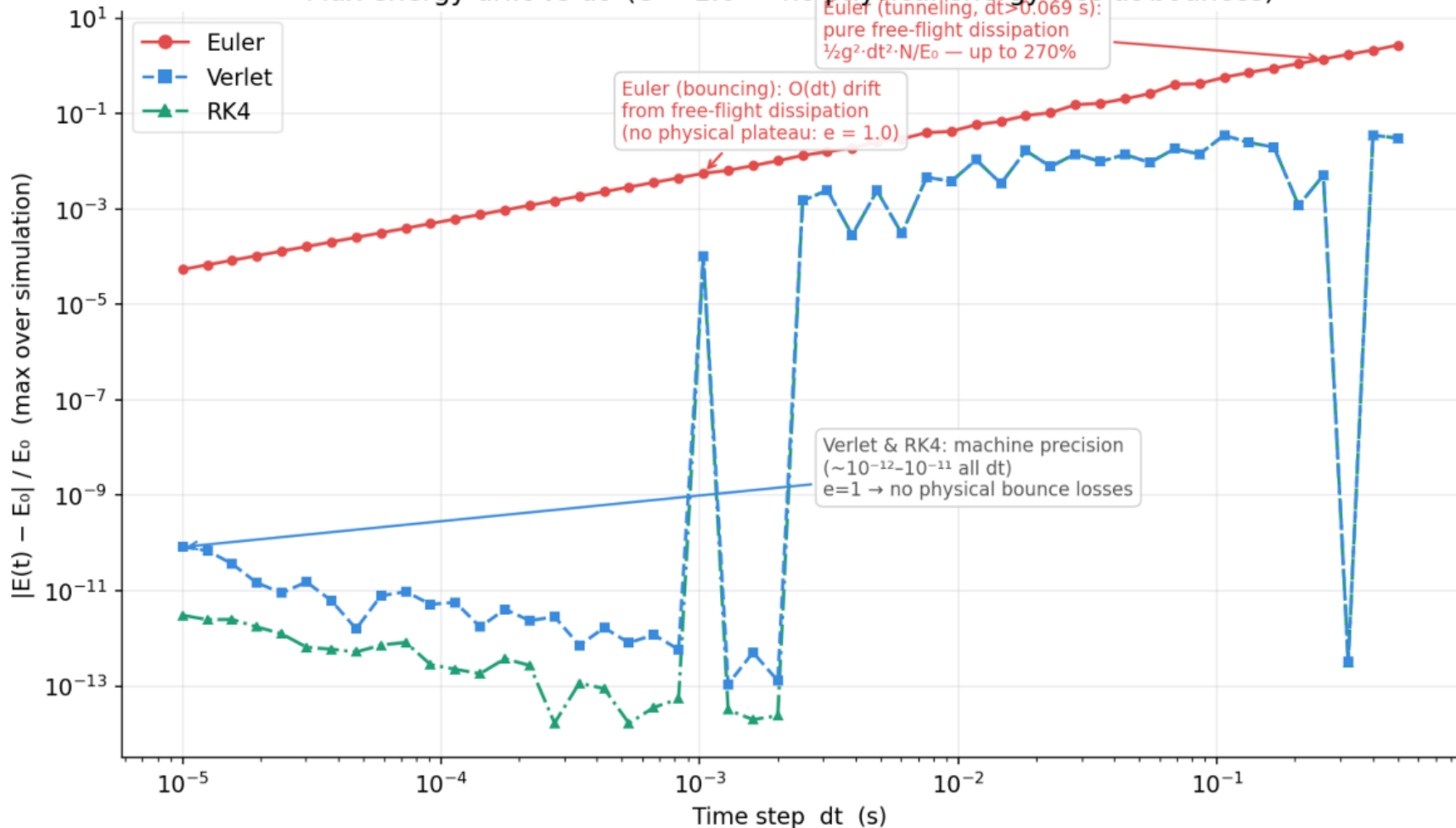


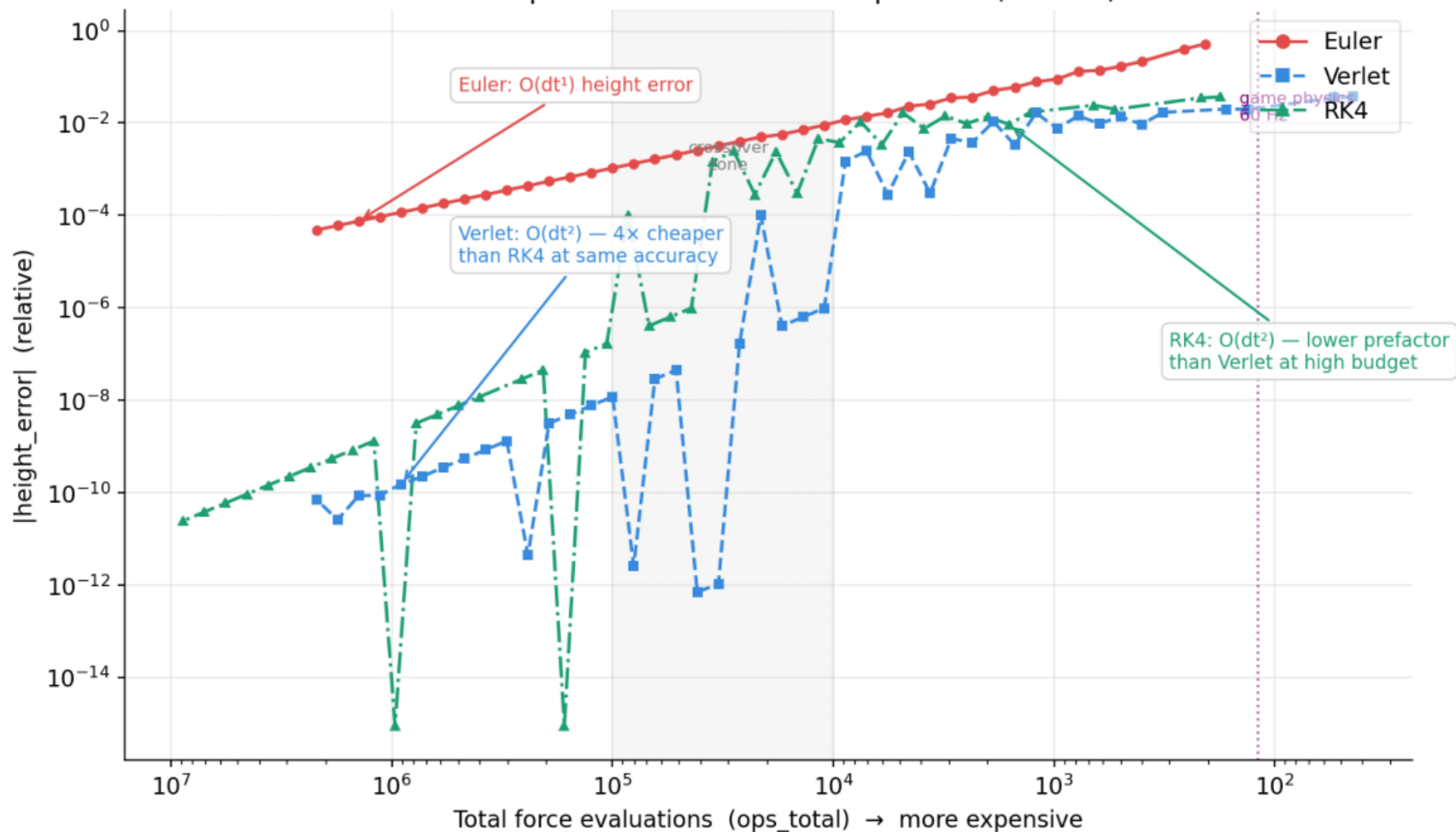
Peak height error vs dt — convergence order (e = 1.0, conservative)

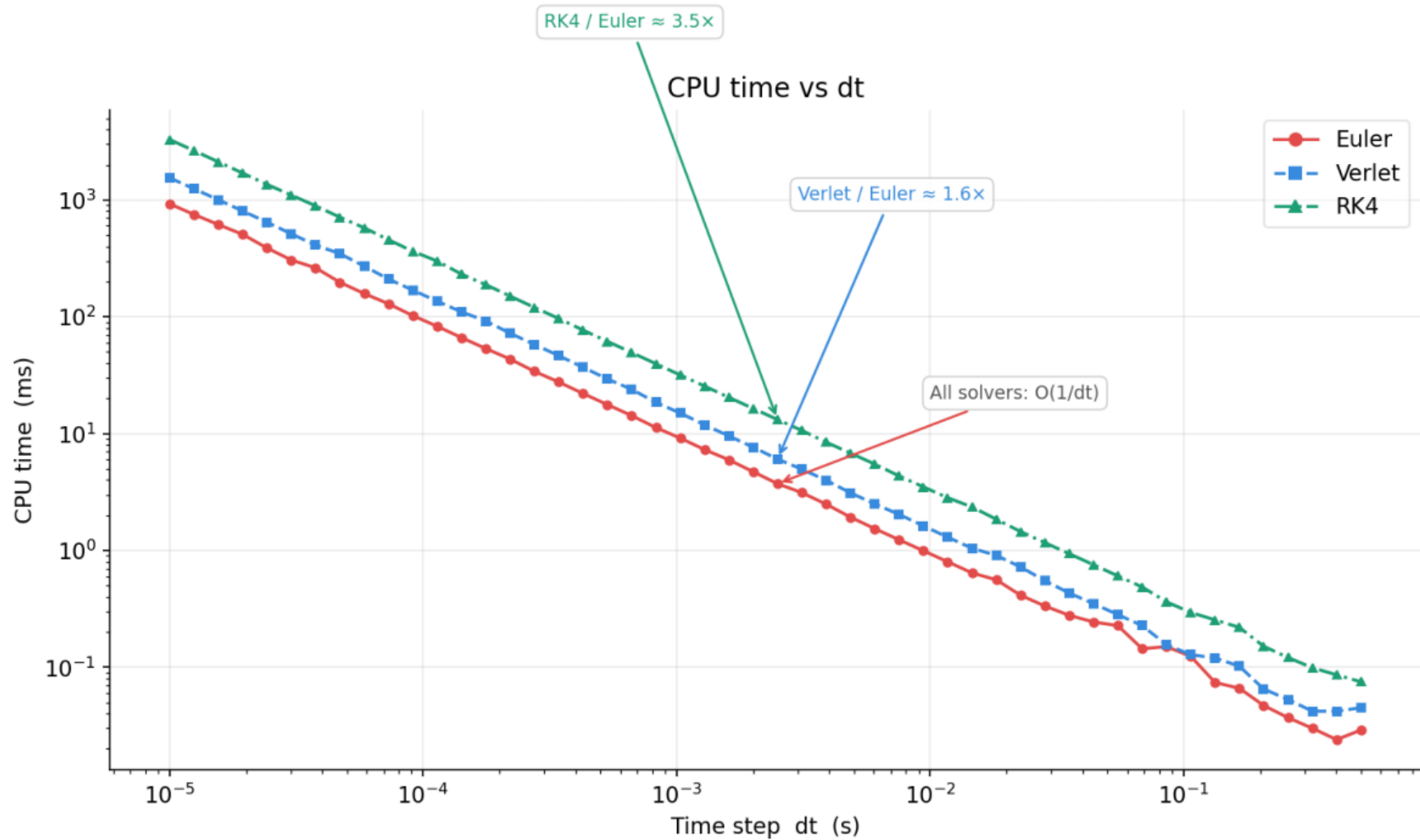


Max energy drift vs dt (e = 1.0 — no physical energy loss at bounces)

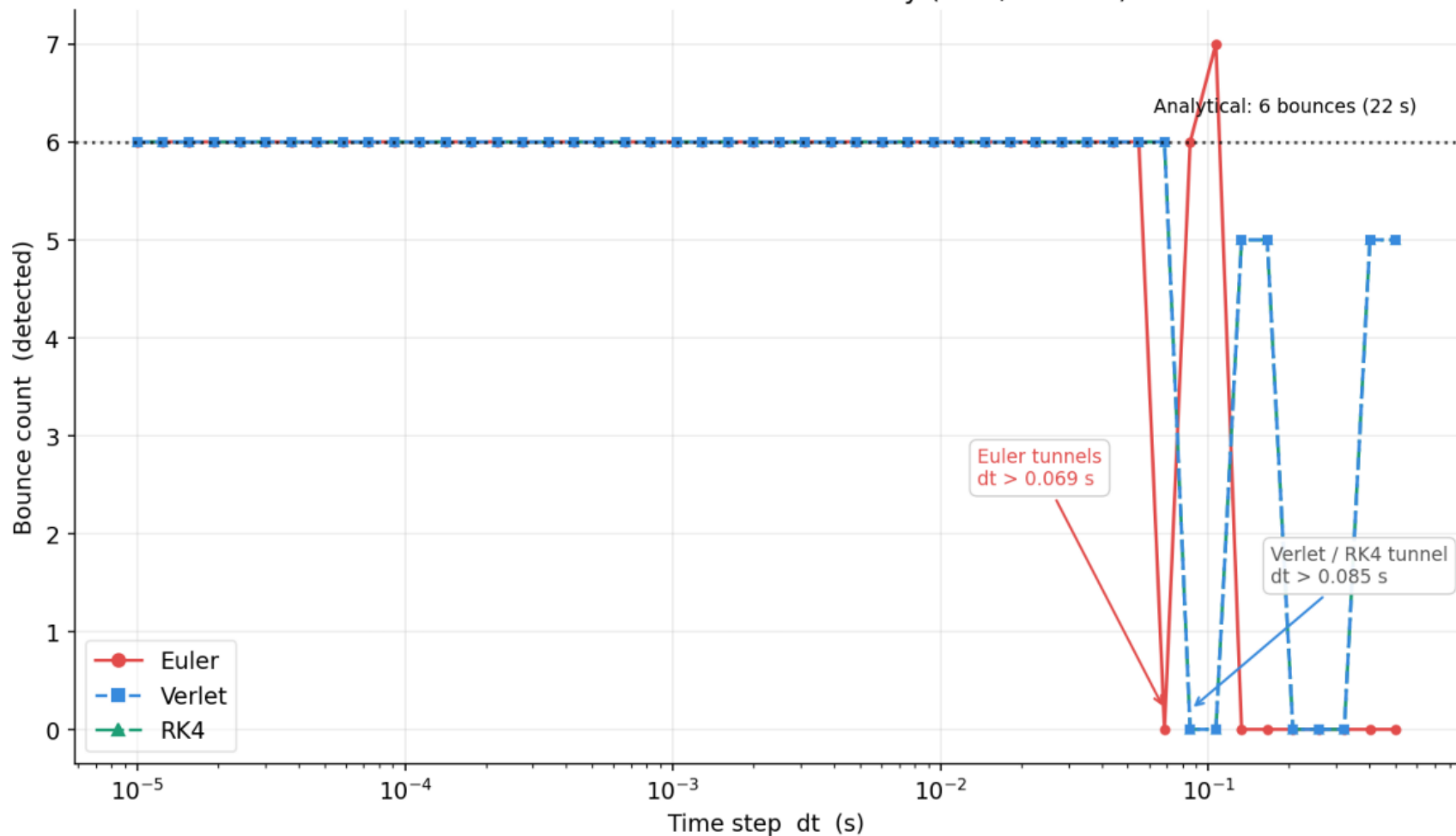


Cost vs precision — iso-cost comparison ($e = 1.0$)





Bounce count vs dt — solver stability (22 s, e = 1.0)



E(t) time series — energy drift over simulation ($e = 1.0$, conservative)

